

Student Academic Performance Analysis Using R

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1. Introduction

Student academic performance is influenced by several factors such as study hours, attendance, assignment performance, internal marks, and examination marks.

This project analyzes student academic performance using R. The analysis focuses on study hours, attendance, assignment scores, internal marks, exam marks, and final scores.

Objectives

- Create and analyze student academic performance data
- Check the data
- Perform statistical analysis
- Create graphs
- Study relationships between variables
- Interpret the results
- Identify important findings
- Provide limitations, future scope, and conclusion

2. Project Background

Academic performance is an important factor in understanding students' learning outcomes.

Data analysis can be used to identify patterns between study habits, attendance, assignments, and examination performance.

R provides useful functions for creating datasets, calculating statistics, analyzing relationships, and creating visualizations.

Scope

This project uses data from 20 sample students.

The main variables are:

- Age
- Study Hours

- Attendance
- Assignment Score
- Internal Marks
- Exam Marks
- Final Score

3. Create Student Academic Performance Data

```
student <- data.frame(  
  Age = c(18,19,20,21,19,22,20,18,21,23,  
          19,20,22,18,21,20,19,23,22,20),  
  
  Study_Hours = c(2.5,3.0,4.0,5.0,3.5,4.5,2.0,5.5,4.0,3.0,  
                  4.5,3.5,5.0,6.0,4.0,2.5,5.5,6.5,3.5,4.5),  
  
  Attendance = c(72,78,85,92,80,88,65,95,87,75,  
                 90,82,94,98,86,70,93,96,79,89),  
  
  Assignment_Score = c(65,70,78,88,72,82,60,92,80,68,  
                       85,75,90,95,79,62,91,96,73,84),  
  
  Internal_Marks = c(58,62,70,82,65,75,55,85,72,60,  
                    80,68,84,90,74,57,86,92,64,78),  
  
  Exam_Marks = c(60,65,72,85,68,78,58,88,75,63,  
                 82,70,87,93,77,61,89,95,67,80),  
  
  Final_Score = c(61,66,73,85,69,79,58,89,76,64,  
                  83,71,88,94,78,62,90,96,68,81)  
)  
  
student
```

```
##      Age Study_Hours Attendance Assignment_Score Internal_Marks Exam_Marks
## 1    18          2.5         72              65          58          60
## 2    19          3.0         78              70          62          65
## 3    20          4.0         85              78          70          72
## 4    21          5.0         92              88          82          85
## 5    19          3.5         80              72          65          68
## 6    22          4.5         88              82          75          78
## 7    20          2.0         65              60          55          58
## 8    18          5.5         95              92          85          88
## 9    21          4.0         87              80          72          75
## 10   23          3.0         75              68          60          63
## 11   19          4.5         90              85          80          82
## 12   20          3.5         82              75          68          70
## 13   22          5.0         94              90          84          87
## 14   18          6.0         98              95          90          93
## 15   21          4.0         86              79          74          77
## 16   20          2.5         70              62          57          61
## 17   19          5.5         93              91          86          89
## 18   23          6.5         96              96          92          95
## 19   22          3.5         79              73          64          67
## 20   20          4.5         89              84          78          80
##      Final_Score
## 1              61
## 2              66
## 3              73
## 4              85
## 5              69
## 6              79
## 7              58
## 8              89
## 9              76
## 10             64
## 11             83
## 12             71
## 13             88
## 14             94
## 15             78
## 16             62
## 17             90
## 18             96
## 19             68
## 20             81
```

4. Dataset Description

The dataset contains 20 student records.

- Age: Age of the student
- Study Hours: Average daily study hours
- Attendance: Attendance percentage
- Assignment Score: Assignment score

- Internal Marks: Internal examination marks
- Exam Marks: Examination marks
- Final Score: Overall academic score

5. Data Checking

```
str(student)
```

```
## 'data.frame': 20 obs. of 7 variables:
## $ Age : num 18 19 20 21 19 22 20 18 21 23 ...
## $ Study_Hours : num 2.5 3 4 5 3.5 4.5 2 5.5 4 3 ...
## $ Attendance : num 72 78 85 92 80 88 65 95 87 75 ...
## $ Assignment_Score: num 65 70 78 88 72 82 60 92 80 68 ...
## $ Internal_Marks : num 58 62 70 82 65 75 55 85 72 60 ...
## $ Exam_Marks : num 60 65 72 85 68 78 58 88 75 63 ...
## $ Final_Score : num 61 66 73 85 69 79 58 89 76 64 ...
```

```
summary(student)
```

```
##      Age      Study_Hours      Attendance      Assignment_Score
## Min.   :18.00   Min.   :2.000   Min.   :65.00   Min.   :60.00
## 1st Qu.:19.00   1st Qu.:3.375   1st Qu.:78.75   1st Qu.:71.50
## Median :20.00   Median :4.000   Median :86.50   Median :79.50
## Mean   :20.25   Mean   :4.125   Mean   :84.70   Mean   :79.25
## 3rd Qu.:21.25   3rd Qu.:5.000   3rd Qu.:92.25   3rd Qu.:88.50
## Max.   :23.00   Max.   :6.500   Max.   :98.00   Max.   :96.00
## Internal_Marks Exam_Marks      Final_Score
## Min.   :55.00   Min.   :58.00   Min.   :58.00
## 1st Qu.:63.50   1st Qu.:66.50   1st Qu.:67.50
## Median :73.00   Median :76.00   Median :77.00
## Mean   :72.85   Mean   :75.65   Mean   :76.55
## 3rd Qu.:82.50   3rd Qu.:85.50   3rd Qu.:85.75
## Max.   :92.00   Max.   :95.00   Max.   :96.00
```

Missing Values

```
colSums(is.na(student))
```

```
##      Age      Study_Hours      Attendance Assignment_Score
##      0           0           0           0
## Internal_Marks      Exam_Marks      Final_Score
##      0           0           0
```

Duplicate Records

```
sum(duplicated(student))
```

```
## [1] 0
```

6. Statistical Analysis

Average Study Hours

```
mean(student$Study_Hours)
```

```
## [1] 4.125
```

Average Attendance

```
mean(student$Attendance)
```

```
## [1] 84.7
```

Average Assignment Score

```
mean(student$Assignment_Score)
```

```
## [1] 79.25
```

Average Internal Marks

```
mean(student$Internal_Marks)
```

```
## [1] 72.85
```

Average Exam Marks

```
mean(student$Exam_Marks)
```

```
## [1] 75.65
```

Average Final Score

```
mean(student$Final_Score)
```

```
## [1] 76.55
```

7. Highest and Lowest Final Score

Maximum Final Score

```
max(student$Final_Score)
```

```
## [1] 96
```

Minimum Final Score

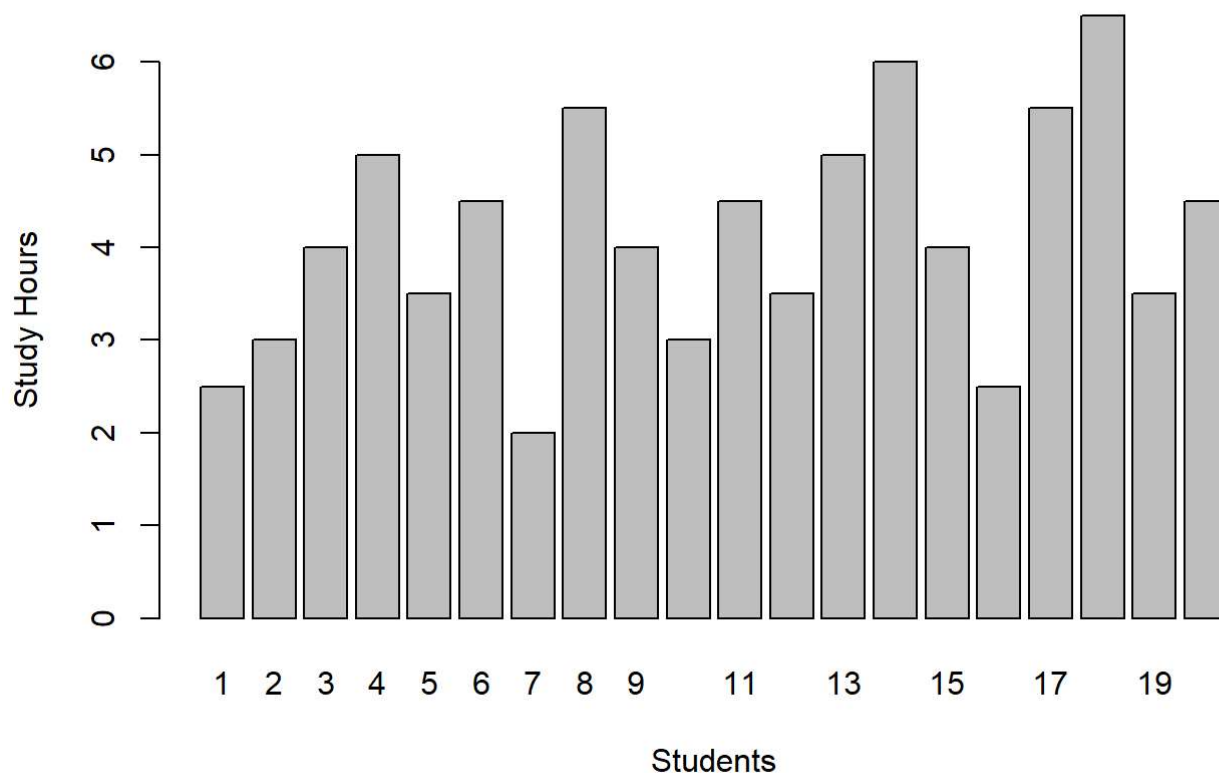
```
min(student$Final_Score)
```

```
## [1] 58
```

8. Study Hours Analysis

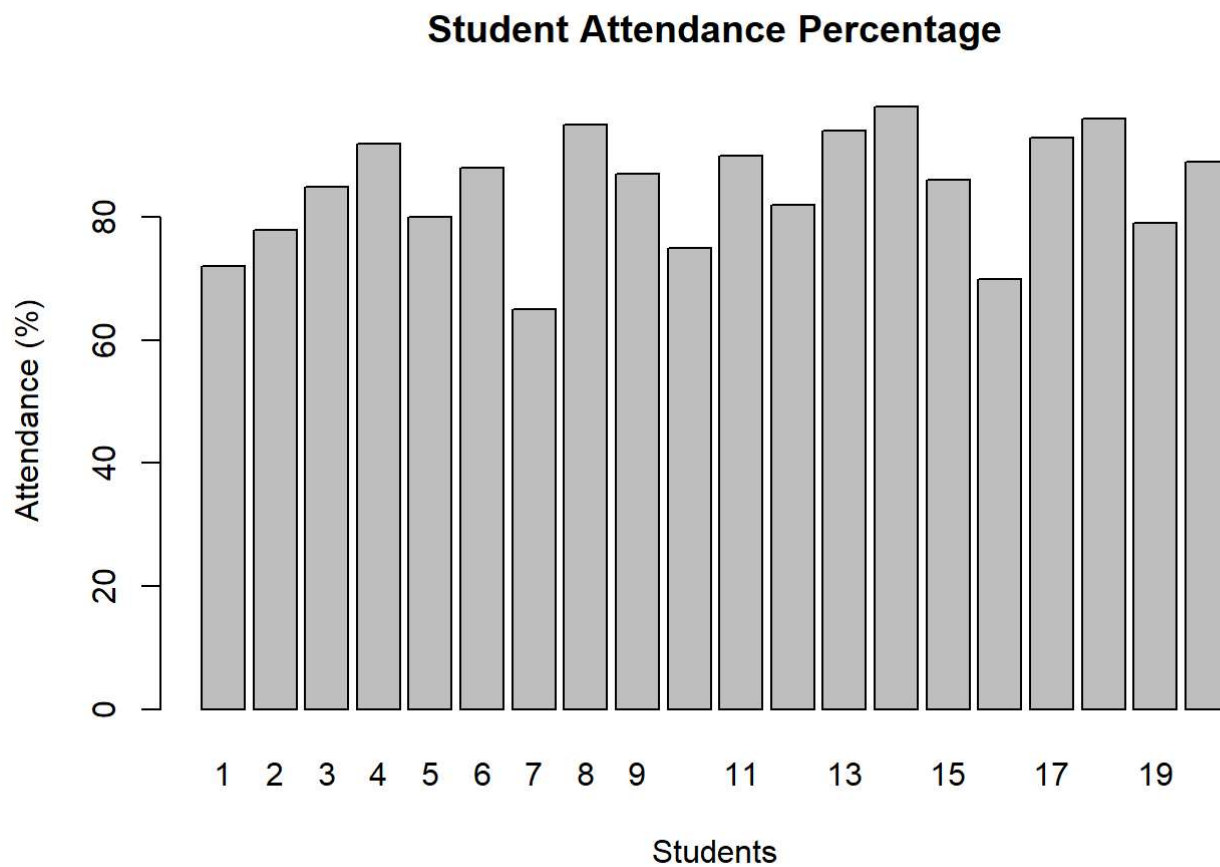
```
barplot(  
  student$Study_Hours,  
  main = "Daily Study Hours of Students",  
  xlab = "Students",  
  ylab = "Study Hours",  
  names.arg = 1:nrow(student)  
)
```

Daily Study Hours of Students



9. Attendance Analysis

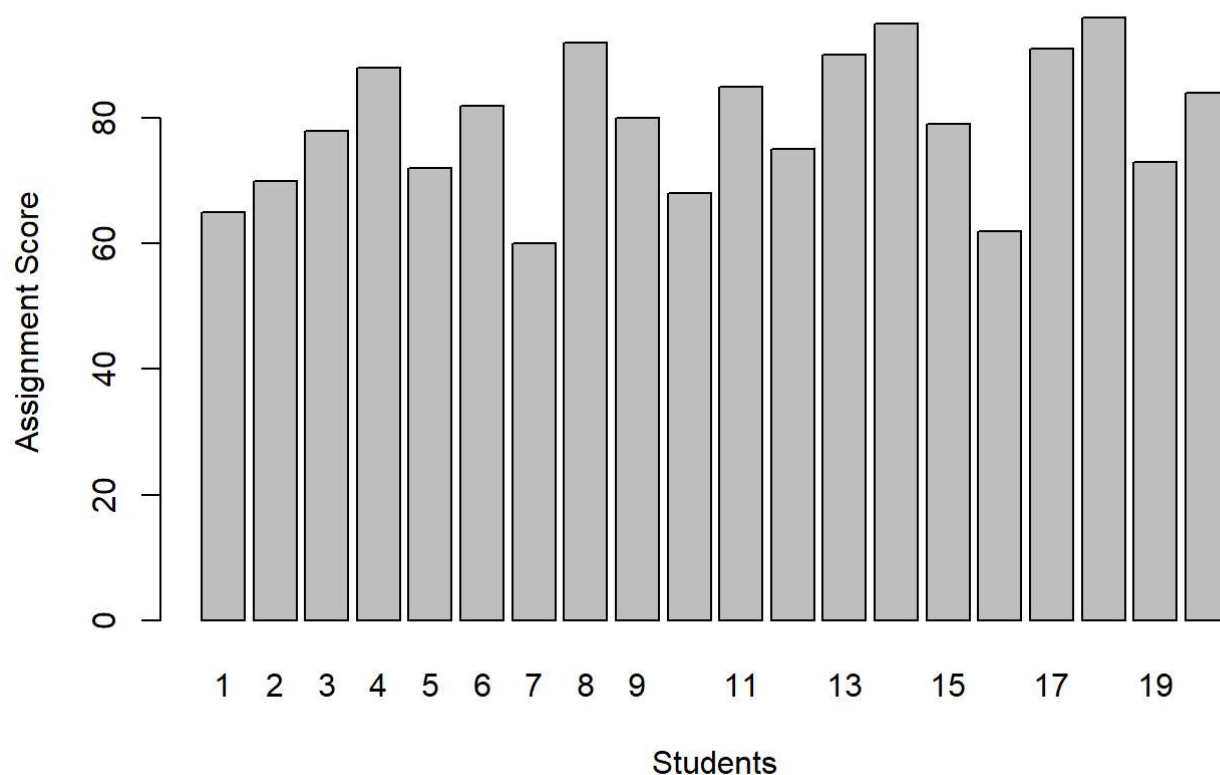
```
barplot(  
  student$Attendance,  
  main = "Student Attendance Percentage",  
  xlab = "Students",  
  ylab = "Attendance (%)",  
  names.arg = 1:nrow(student)  
)
```



10. Assignment Score Analysis

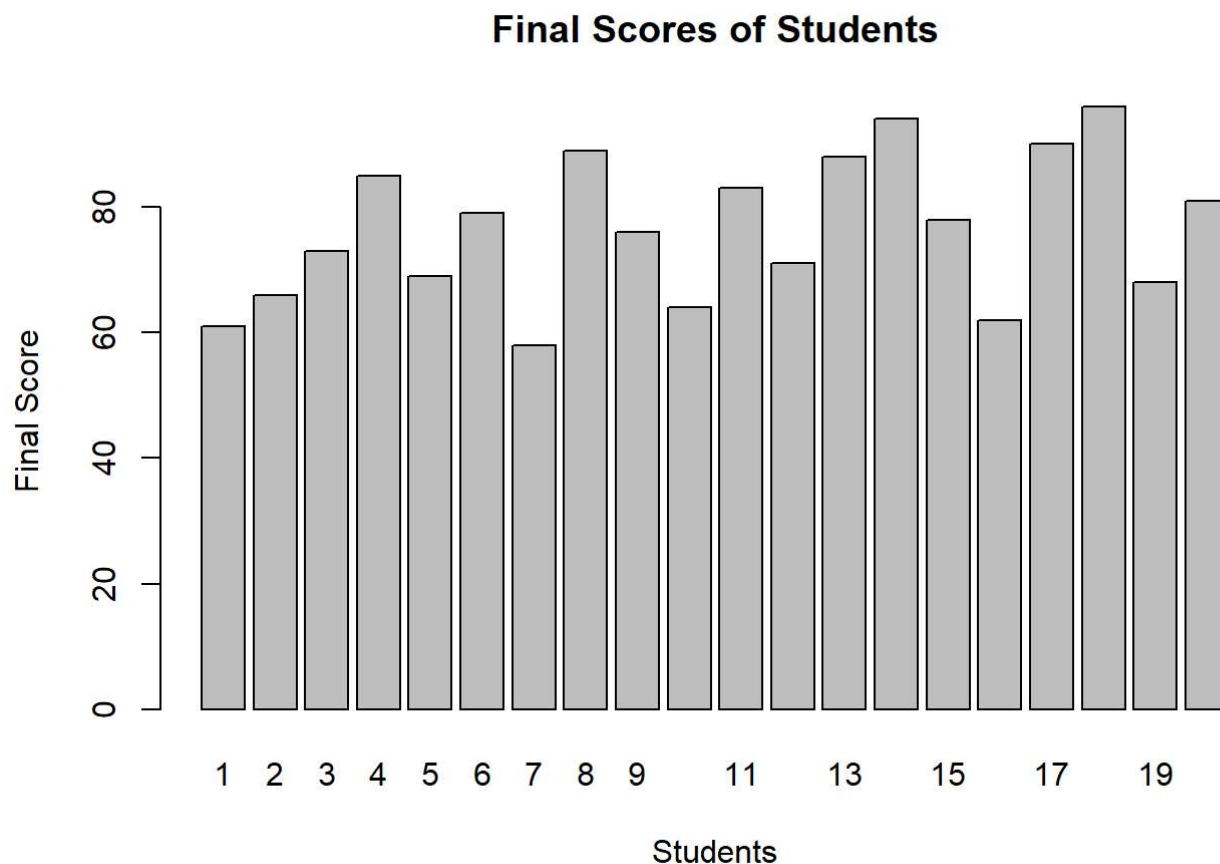
```
barplot(  
  student$Assignment_Score,  
  main = "Assignment Scores of Students",  
  xlab = "Students",  
  ylab = "Assignment Score",  
  names.arg = 1:nrow(student)  
)
```

Assignment Scores of Students



11. Final Score Analysis

```
barplot(  
  student$Final_Score,  
  main = "Final Scores of Students",  
  xlab = "Students",  
  ylab = "Final Score",  
  names.arg = 1:nrow(student)  
)
```

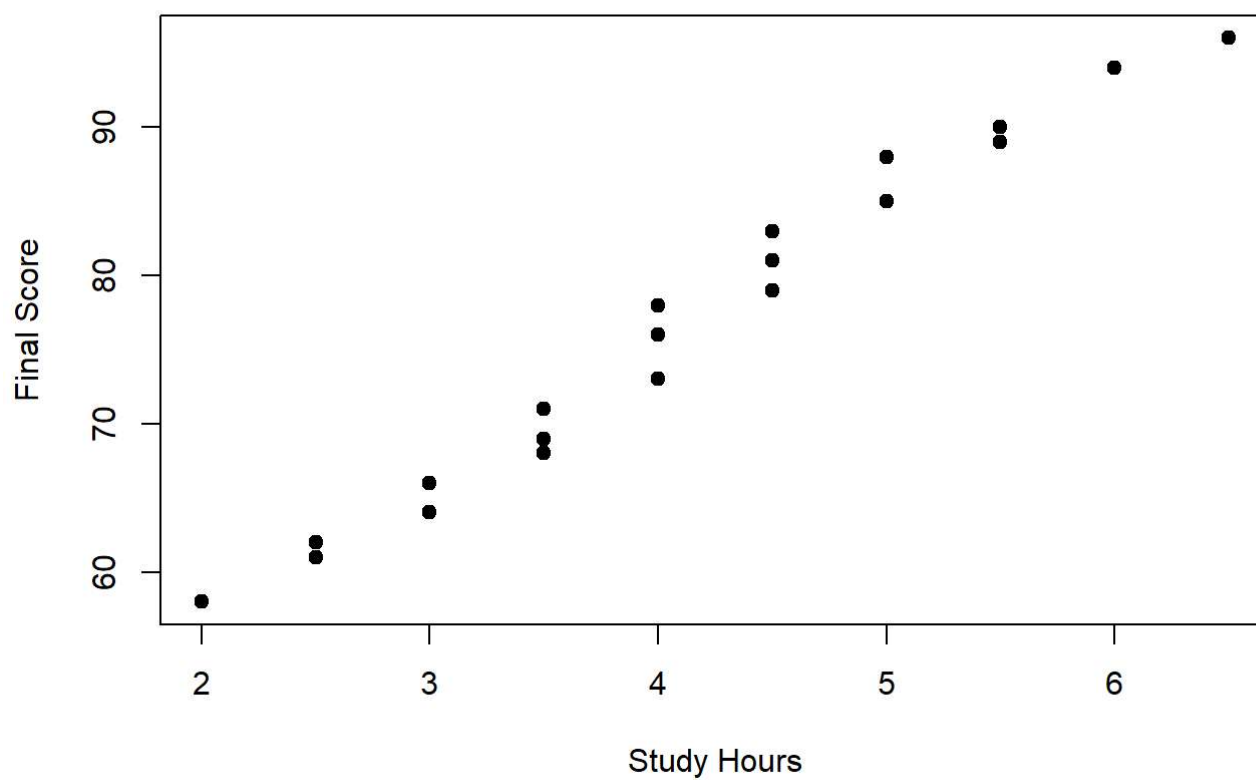


12. Relationship Analysis

Study Hours vs Final Score

```
plot(  
  student$Study_Hours,  
  student$Final_Score,  
  main = "Study Hours vs Final Score",  
  xlab = "Study Hours",  
  ylab = "Final Score",  
  pch = 19  
)
```

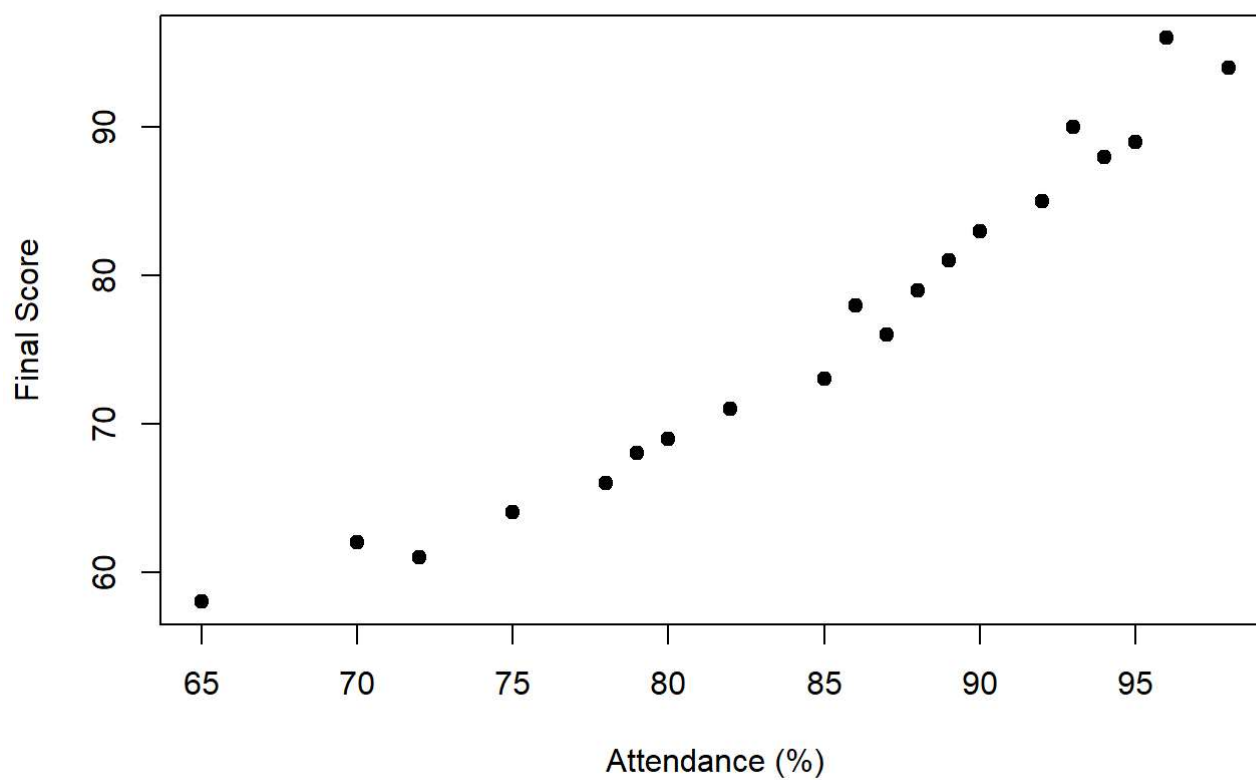
Study Hours vs Final Score



Attendance vs Final Score

```
plot(  
  student$Attendance,  
  student$Final_Score,  
  main = "Attendance vs Final Score",  
  xlab = "Attendance (%)",  
  ylab = "Final Score",  
  pch = 19  
)
```

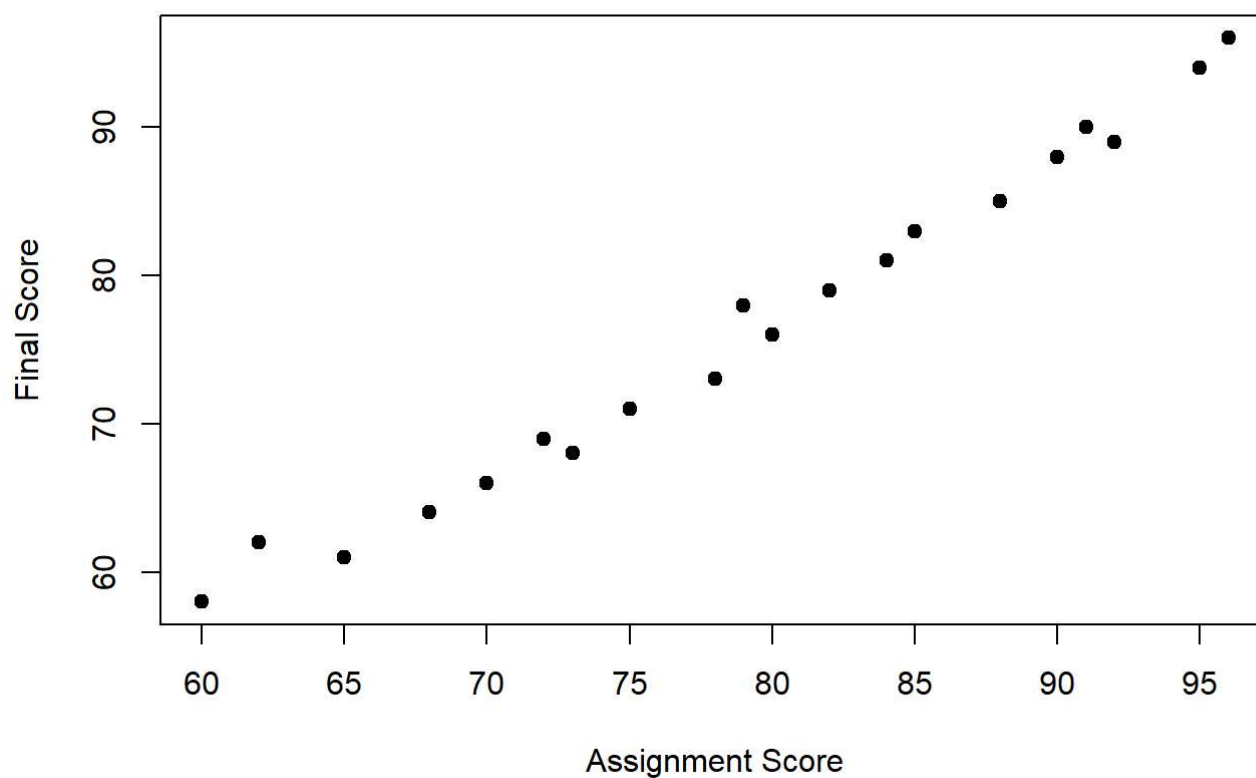
Attendance vs Final Score



Assignment Score vs Final Score

```
plot(  
  student$Assignment_Score,  
  student$Final_Score,  
  main = "Assignment Score vs Final Score",  
  xlab = "Assignment Score",  
  ylab = "Final Score",  
  pch = 19  
)
```

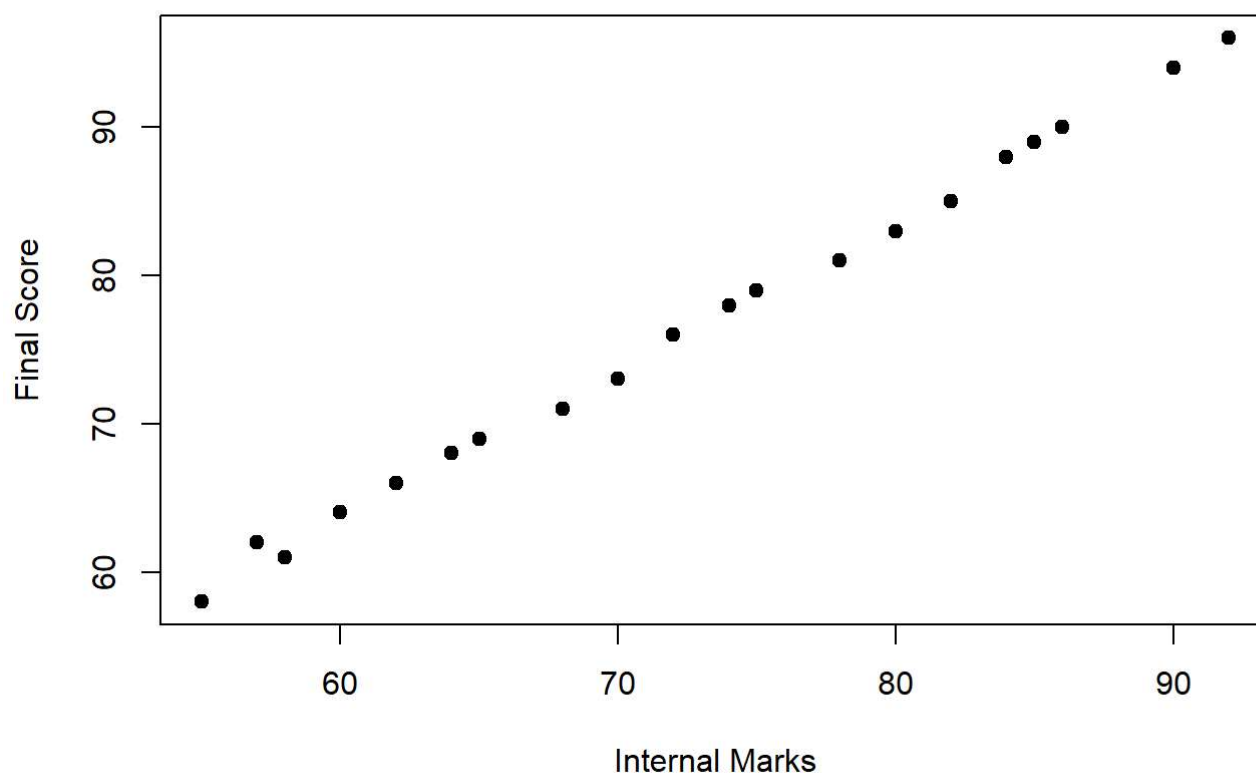
Assignment Score vs Final Score



Internal Marks vs Final Score

```
plot(  
  student$Internal_Marks,  
  student$Final_Score,  
  main = "Internal Marks vs Final Score",  
  xlab = "Internal Marks",  
  ylab = "Final Score",  
  pch = 19  
)
```

Internal Marks vs Final Score



13. Correlation Analysis

Study Hours and Final Score

```
cor(  
  student$Study_Hours,  
  student$Final_Score  
)
```

```
## [1] 0.9882417
```

Attendance and Final Score

```
cor(  
  student$Attendance,  
  student$Final_Score  
)
```

```
## [1] 0.9733975
```

Assignment Score and Final Score

```
cor(  
  student$Assignment_Score,  
  student$Final_Score  
)
```

```
## [1] 0.9921657
```

Internal Marks and Final Score

```
cor(  
  student$Internal_Marks,  
  student$Final_Score  
)
```

```
## [1] 0.9987752
```

Exam Marks and Final Score

```
cor(  
  student$Exam_Marks,  
  student$Final_Score  
)
```

```
## [1] 0.9996498
```

14. Complete Correlation Matrix

```
cor(student)
```

```
##           Age Study_Hours Attendance Assignment_Score
## Age      1.00000000 0.06445123 0.01959331      0.03258203
## Study_Hours 0.06445123 1.00000000 0.96594293      0.98726278
## Attendance  0.01959331 0.96594293 1.00000000      0.98912209
## Assignment_Score 0.03258203 0.98726278 0.98912209      1.00000000
## Internal_Marks 0.01946443 0.98600076 0.97422110      0.99356182
## Exam_Marks   0.03103591 0.98717735 0.97086547      0.99140479
## Final_Score  0.02948461 0.98824174 0.97339746      0.99216569
##           Internal_Marks Exam_Marks Final_Score
## Age      0.01946443 0.03103591 0.02948461
## Study_Hours 0.98600076 0.98717735 0.98824174
## Attendance 0.97422110 0.97086547 0.97339746
## Assignment_Score 0.99356182 0.99140479 0.99216569
## Internal_Marks 1.00000000 0.99896714 0.99877517
## Exam_Marks   0.99896714 1.00000000 0.99964976
## Final_Score  0.99877517 0.99964976 1.00000000
```

15. Interpretation and Findings

Interpretation

The analysis provides several observations:

- Students have different daily study hours.
- Attendance percentages vary among students.
- Assignment scores differ between students.
- Internal marks show variation.
- Examination marks vary among students.
- Final scores are different for different students.
- Study hours can be compared with final academic scores.
- Attendance can be compared with final academic scores.
- Assignment performance can be compared with final scores.
- Internal marks can be compared with final scores.

These observations are based only on the sample dataset used in this project.

Key Findings

- The dataset contains 20 student records.
- There are 7 variables.
- The dataset can be checked for missing values and duplicate records.
- Students have different study patterns.
- Attendance varies among students.
- Assignment performance varies among students.
- Internal and examination marks vary among students.
- Final academic scores differ among students.
- R can be used effectively for academic data analysis and visualization.

16. Limitations

This project uses a small sample of 20 students.

The dataset does not include every factor that may influence academic performance.

Factors such as teaching methods, learning environment, extracurricular activities, and personal study preferences are not included.

Student performance can also change over time.

17. Future Scope

The project can be improved by adding:

- More students
- Multiple semesters
- Subject-wise marks
- Attendance for individual subjects
- Assignment submission data
- Study methods
- Previous semester CGPA
- Online learning hours
- Extracurricular activities
- Revision hours

A larger dataset could provide more detailed academic analysis.

18. Conclusion

This project analyzed student academic performance using R.

It demonstrated how to create an academic dataset, check data quality, calculate statistical measures, create graphs, analyze relationships, and calculate correlations.

The project provides an understanding of student academic performance using data analysis and visualization.